

Laplace's Equation, Uniqueness, and Images

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Presentation map

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Source

These slides summarize the electrostatics logic developed in Griffiths, *Introduction to Electrodynamics*, 4th ed., Chapter 3, especially Sections 3.1 and 3.2.

Why Laplace's Equation Matters

Electrostatic potentials are governed by Poisson's equation,

$$\nabla^2 V(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon_0}.$$

In any charge-free region, $\rho = 0$, and therefore

$$\boxed{\nabla^2 V = 0.}$$

This is Laplace's equation. It does not mean the potential is trivial; it means the local value of V is determined entirely by surrounding boundary data.

Griffiths viewpoint

Electrostatics is fundamentally a boundary-value theory: sources matter, but the physically realized solution is selected by the boundaries.

One-Dimensional Warm-Up

If the potential depends only on one coordinate, Laplace's equation becomes

$$\frac{d^2V}{dx^2} = 0.$$

Integrating twice gives

$$V(x) = C_1x + C_2.$$

So in one dimension, a charge-free potential is linear. The constants are not fixed by the differential equation itself; they come from boundary conditions such as

$$V(0) = V_0, \quad V(a) = V_a.$$

Key lesson

Even the simplest harmonic solution shows that local source-free behavior and global boundary data are inseparable.

Harmonic Functions and the Averaging Property

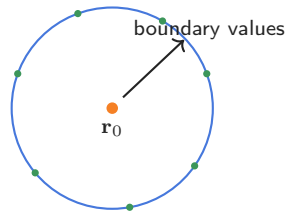
Solutions of Laplace's equation are called *harmonic functions*. Griffiths emphasizes a remarkable property:

$$V(\mathbf{r}_0) = \text{average of } V \text{ over any sphere centered at } \mathbf{r}_0$$

provided the sphere lies entirely in a charge-free region.

This means a harmonic function cannot be shaped by local spikes or dips. The value at a point is tied to the values all around it.

It also explains why field-free regions can still have nontrivial potentials: outside charges and boundaries determine the average.



center value equals spherical average

No Local Extrema in Empty Space

The averaging property has an immediate consequence:

A harmonic function cannot have a local maximum or minimum in the interior.

If an interior point were a true maximum, the average over a small sphere around it would have to be smaller than the center value, contradicting the averaging theorem. The same reasoning rules out a local minimum.

Physical interpretation

In a charge-free region, the potential cannot “peak” or “bottom out” by itself. Any extremum must be imposed by the boundary.

First Uniqueness Theorem

Suppose the charge density $\rho(\mathbf{r})$ is known throughout a volume and the potential is specified on the entire boundary. Then the electrostatic solution is unique.

The practical consequence is stronger than it first appears:

- if you find *any* function satisfying Poisson's equation in the volume,
- and it matches the boundary potential,
- then it is automatically the physical answer.

Why this matters

This theorem turns clever guesses into rigorous solutions.

Conductors and the Second Uniqueness Theorem

In electrostatic equilibrium, each conductor is an equipotential surface, but those constant values may not be known in advance. Griffiths reformulates the problem this way:

If the total charge on each conductor is specified, together with the volume charge density ρ , then the electric field is uniquely determined.

- The surface charge distribution is not known beforehand.
- The constant potential on each conductor is not known beforehand.
- Yet the final field configuration is still unique once the total conductor charges are fixed.

Use in practice

This theorem is what lets conductor problems be solved by indirect constructions, not by brute-force guessing of surface charge densities.

Boundary-Value Strategy in Electrostatics

Griffiths's Chapter 3 logic can be summarized as a workflow:

- ① identify the region where you want the potential,
- ② write the governing equation there,
- ③ encode the physical boundary conditions,
- ④ construct a candidate solution using symmetry or an auxiliary problem,
- ⑤ invoke uniqueness to certify it.

Core idea

The hard part is often not solving the PDE directly. The hard part is finding the right representation that already respects the boundaries.

Classic Image Problem: Charge Above a Grounded Plane

A point charge q sits at height d above an infinite grounded conducting plane. In the region $z > 0$, the boundary conditions are

$$V = 0 \quad \text{at } z = 0, \quad V \rightarrow 0 \quad \text{far away.}$$

Replace the conductor by an *image charge* $-q$ at $(0, 0, -d)$. Then for $z \geq 0$,

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right]$$

Reason it works

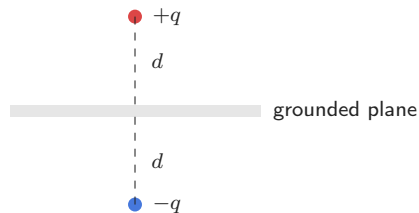
This auxiliary two-charge problem satisfies the correct differential equation in $z > 0$ and forces $V = 0$ on the plane, so uniqueness says it must equal the conductor solution there.

Image-Charge Geometry

The image charge is not physically present. It is placed only to make the boundary condition automatic.

For a grounded plane:

- the real charge is $+q$ at $z = d$,
- the image charge is $-q$ at $z = -d$,
- points on the plane are equidistant from both charges,
- so the two contributions cancel there.



Induced Surface Charge on the Plane

Once the potential is known, the induced surface charge follows from the normal derivative:

$$\sigma = -\epsilon_0 \left. \frac{\partial V}{\partial n} \right|_{\text{surface}}.$$

For the grounded plane, the normal direction is $\hat{\mathbf{z}}$, so

$$\sigma(x, y) = -\epsilon_0 \left. \frac{\partial V}{\partial z} \right|_{z=0} = \boxed{-\frac{qd}{2\pi(x^2 + y^2 + d^2)^{3/2}}}.$$

The total induced charge is

$$Q_{\text{ind}} = \int \sigma \, da = -q.$$

Interpretation

The conductor rearranges exactly enough negative charge to terminate the electric field on its surface while keeping the plane at zero potential.

Force and Energy: A Useful Caution

Near the real charge, the field in the physical problem matches the field in the auxiliary two-charge problem, so the force on q is

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{\mathbf{z}}.$$

But the *energy* is not the same as the energy of two actual point charges in free space. For the grounded-plane problem,

$$W = \boxed{-\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}}.$$

This is one-half of the two-point-charge value because only the upper half-space contains field energy in the conductor problem.

Lesson

The image construction reproduces the potential in the region of interest, but it is not a license to identify every global quantity with the auxiliary problem.

Grounded Conducting Sphere

For a point charge q outside a grounded conducting sphere of radius R , Griffiths gives another exact image construction.

If the real charge is a distance a from the center ($a > R$), then place an image charge

$$q' = -\frac{R}{a}q$$

at distance

$$b = \frac{R^2}{a}$$

from the center along the same axis, inside the sphere.

The outside potential is then

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} + \frac{q'}{r'} \right)$$

where r and r' are distances from the real and image charges.

Constraint

Image charges must lie outside the region where you are solving Poisson's equation; otherwise you

Main Takeaways from Griffiths Chapter 3

- Laplace's equation describes source-free regions, but the solution is still globally shaped by the boundary.
- Harmonic functions obey an averaging property and therefore cannot have interior extrema.
- Uniqueness theorems justify indirect methods: once the equation and boundaries are matched, the answer is fixed.
- The method of images works by replacing a conductor problem with an auxiliary source configuration that reproduces the same potential in the region of interest.
- Image methods are elegant but geometry-dependent; they succeed only for specially structured boundaries.

Reference

Based on Griffiths, *Introduction to Electrodynamics*, 4th ed., Chapter 3, Sections 3.1–3.2.